Number System

Data Representation

- makes it possible to convert letters, sounds, and images into a form computers can use for processing.

- Bit Patterns are used to represent all types of data
  - Numbers
  - Text characters
  - Images
  - Sound

Types of number systems

- Decimal number system
- Binary number system
- Octal number system
- Hexadecimal number system

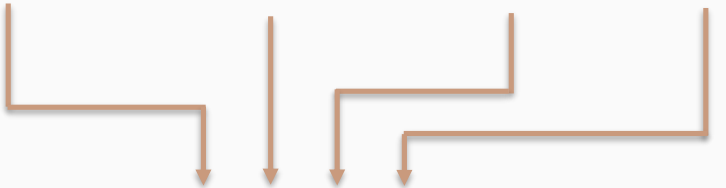
Decimal numbers (Base 10)

- The number system that we use in our day-to-day life is the decimal number system. Decimal number system has base 10 as it uses 10 digits from 0 to 9.

- Each position represents a specific power of the base (10). For example, the decimal number 1234 consists of the digit 4 in the units position, 3 in the tens position, 2 in the hundreds position, and 1 in the thousands position.

$$1x10^3 + 2x10^2 + 3x10^1 + 4x10^0$$

1000 + 200 + 30 + 4



1 2 3 4

Binary number system (Base 2)

- Uses two digits, 0 and 1.
- Also called as base 2 number system.
- $(01010)_2$

Octal number system (Base 8)

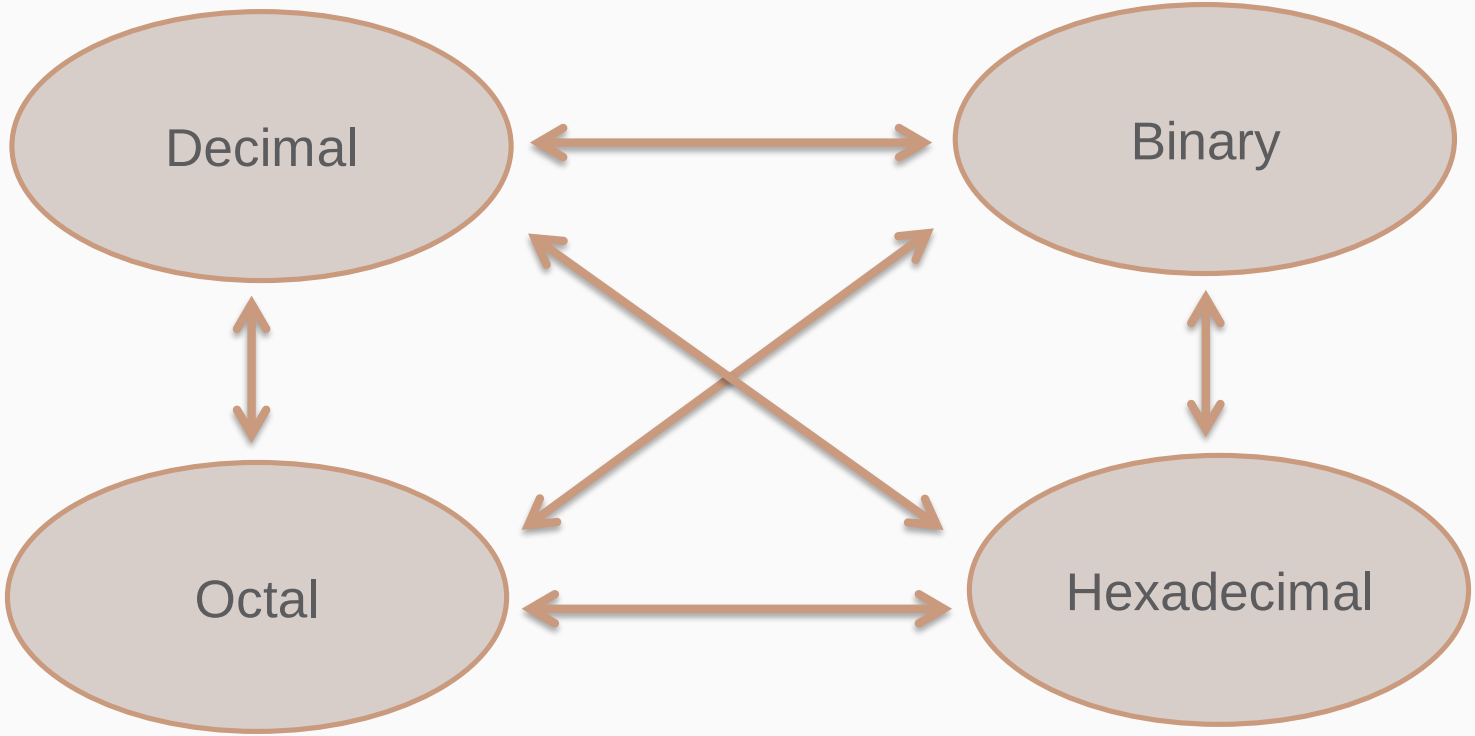
- Uses eight digits, 0, 1, 2, 3, 4, 5, 6, 7
- Also called as base 8 number system.
- $(02571)_8$

Hexadecimal number system (Base 16)

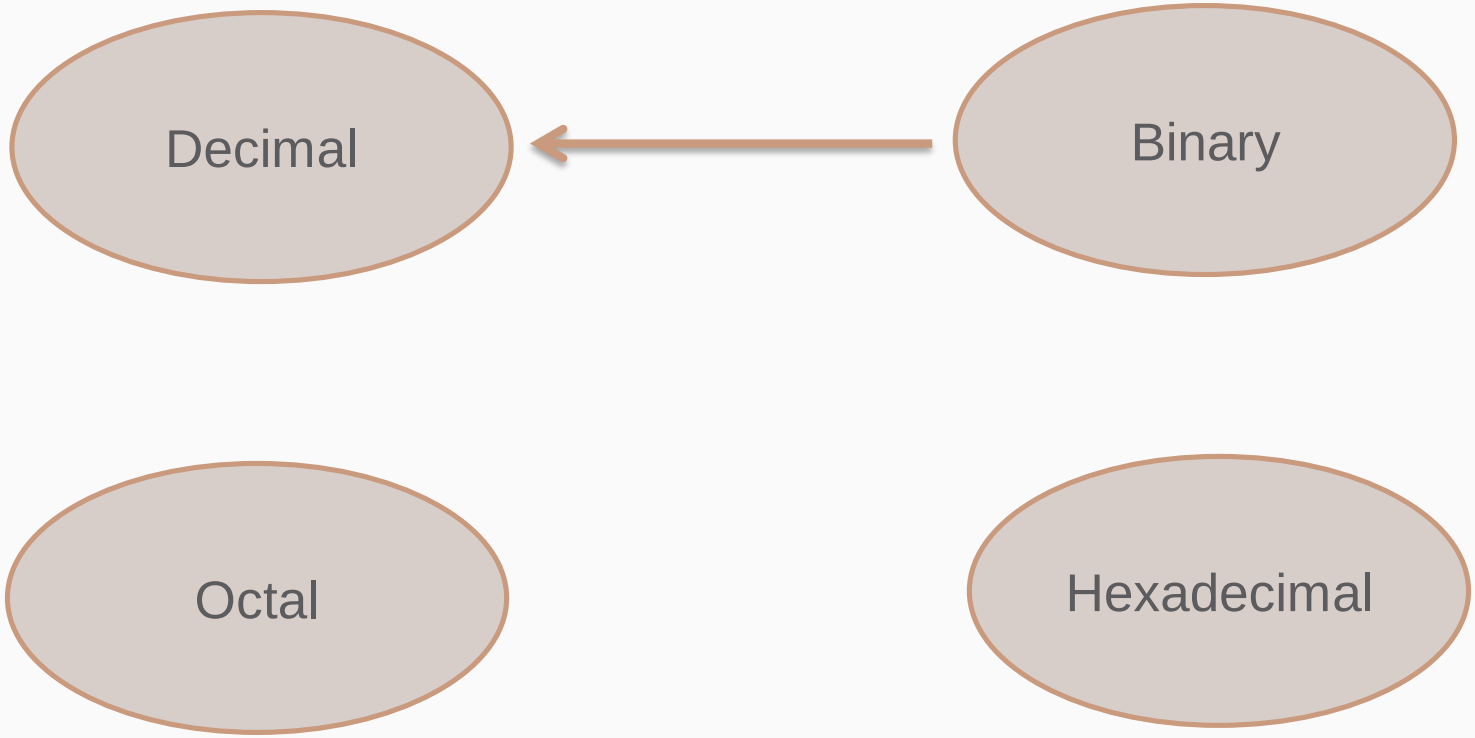
- Uses 10 digits and 6 letters, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F
- Letters represent the numbers starting from 10
- Also called as base 16 number system.
- $(02571)_{16}$

A	10
B	11
C	12
D	13
E	14
F	15

Conversion Among Bases

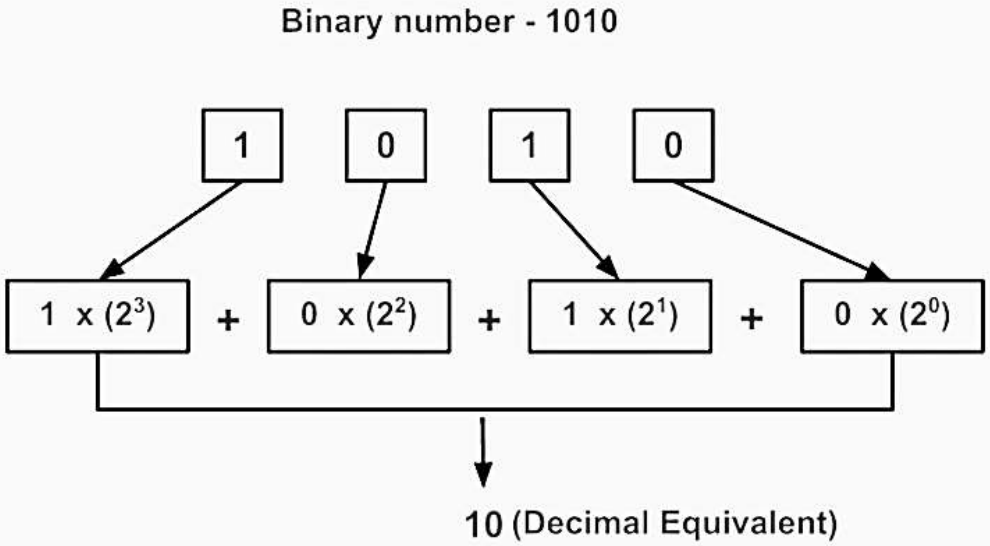


Binary to Decimal



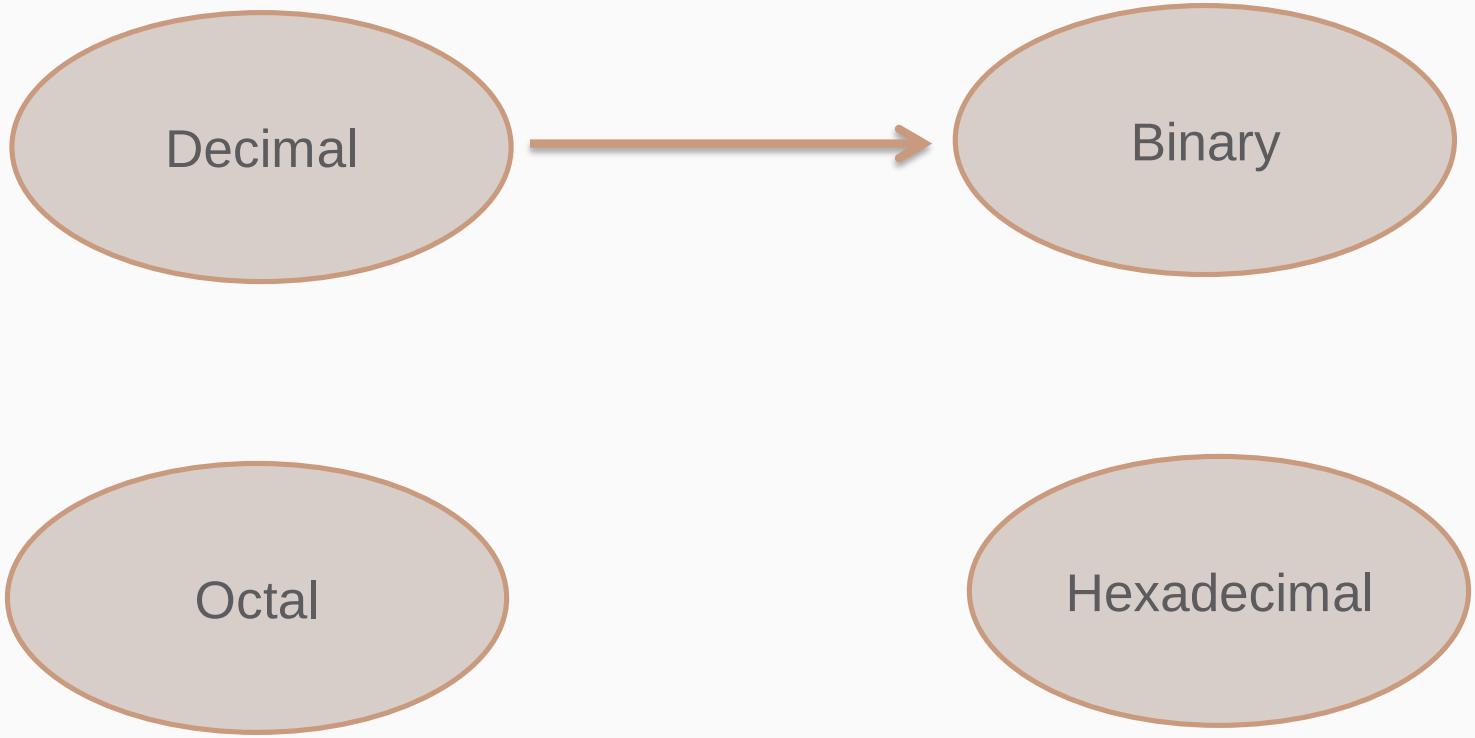
Technique

- Multiply each bit by 2n, where n is the “weight” of the bit
- The weight is the position of the bit, starting from 0 on the right
- Add the results



- Try to solve :
  - $(100011)_2$
  - $(1100101)_2$

Decimal to Binary



Technique

- Divide by two, keep track of the remainder
- First remainder is bit 0 (LSB, least-significant bit)
- Second remainder is bit 1

Given: Decimal Number = $17_{10} = (10001)_2$

Divide by 2	Result	Binary Value (Reminder)
17 / 2	8	1 (LSB)
8 / 2	4	0
4 / 2	2	0
2 / 2	1	0
1 / 2	0	1 (MSB)

- Try to solve :
  - $(34)_{10}$
  - $(160)_{10}$

Thanks!